

# Door Problem Write Up

## Problem

The Door subsystem requires an end-to-end system that takes a **continuous sensor stream, identifies individual door operations**, and classify each operation as **Normal or Abnormal resistance**.

## Data Exploration

The training stream contains 18,036 sensor readings across 17 raw columns. The main physical signals used throughout the work are motor current, motor voltage, motor electrodynamic force, and door leaf position. The controller data also contains command, switch, lock, and operating-state variables.

The training ground truth contains 110 labelled operations. There are 55 Open operations and 55 Close operations, with 80 Normal and 30 Abnormal resistance cycles. The class breakdown is balanced across operation type, with 40 Normal and 15 Abnormal cycles for each of Open and Close.

| Operation | Normal | Abnormal resistance | Total |
|-----------|--------|---------------------|-------|
| Open      | 40     | 15                  | 55    |
| Close     | 40     | 15                  | 55    |
| Total     | 80     | 30                  | 110   |

The custom timestamps are stored as Year-Month-Day-Hour-Minute-Second-Millisecond strings and were converted into Pandas timestamps for time-difference calculations, duration analysis, and submission generation. **No invalid timestamps were found in the analysed training or test streams.**

## Temporal Segmentation

We investigated the temporal structure directly and the key discovery was the difference between within-operation sampling and the gaps between operations. Readings within an operation occur at approximately 20ms intervals (about 50Hz), while consecutive operations are separated by much larger timestamp discontinuities

We then treated a **timestamp gap greater than 1.0s as an operation boundary**, serving as a conservative threshold relative to the observed 20ms sampling interval. This produced 109 internal gaps, which corresponds to exactly 110 labelled operations as N sequential segments will require N-1 internal boundaries. These detected start and end timestamps were then compared against **Train\_Segments\_Answer.csv**:

```
[30]: # Summarise how accurately the timestamp-gap method reproduces the labelled boundaries.
```

```
if len(detected) == len(truth):  
  
    print("Start-time error (ms):")  
    display(segmentation_comparison["start_error_ms"].describe())  
  
    print("\nEnd-time error (ms):")  
    display(segmentation_comparison["end_error_ms"].describe())  
  
    print("\nRow-count error:")  
    display(segmentation_comparison["row_error"].describe())
```

```
Start-time error (ms):  
count    110.0  
mean      0.0  
std       0.0  
min       0.0  
25%      0.0  
50%      0.0  
75%      0.0  
max       0.0  
Name: start_error_ms, dtype: float64
```

```
End-time error (ms):  
count    110.0  
mean      0.0  
std       0.0  
min       0.0  
25%      0.0  
50%      0.0  
75%      0.0  
max       0.0  
Name: end_error_ms, dtype: float64
```

```
Row-count error:  
count    110.0  
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min       0.0  
25%      0.0  
50%      0.0  
75%      0.0  
max       0.0  
Name: row_error, dtype: float64
```

This shows that the timestamp gap method was able to reproduce the provided training segmentation exactly. **For training-cycle construction, the official start\_time and end\_time labels remain the ground-truth source.** The gap-based method is the inference-time mechanism that can be applied when Test labels are unavailable.

### **Fault Analysis**

Once individual cycles were isolated, analysis focused on determining what distinguishes Normal from Abnormal resistance. Open and Close operations were examined separately because their normal operating profiles differ mechanically.

Cycle duration did not show a strong standalone relationship with fault status. Open cycles had essentially identical average durations between Normal and Abnormal groups, while Close cycles showed only a small average difference. Peak motor current also provided limited standalone separation because maximum current values were very similar across classes

Mean motor current was substantially more informative. Abnormal cycles showed higher average current in both Open and Close operations. This suggested that the fault

signature is related to sustained electrical effort and signal behaviour over the course of the operation, rather than a single extreme peak.

Normalised 0-100% cycle profiles were then used to compare signal shape independently of absolute cycle duration. This analysis motivated phase-aware and dynamic feature engineering, while avoiding the assumption that any one sensor threshold is sufficient.

### **Feature Engineering**

Each segmented cycle was converted into a fixed-length feature vector for supervised classification. The initial feature set contained roughly 140 numerical candidates. Feature quality checks found no NaN or infinite values, and two constant features were identified and removed.

| Feature family                | Purpose                                                 |
|-------------------------------|---------------------------------------------------------|
| Whole-cycle statistics        | Capture overall signal level and variability            |
| Phase-based statistics        | Capture where signal changes occur during the operation |
| Dynamic / derivative features | Represent signal and position changes between samples   |
| Movement features             | Describe door-position movement behaviour               |
| Integral-like features        | Approximate accumulated electrical/mechanical effort    |
| Timing                        | Retain cycle duration and sample count                  |
| Operation context             | Encode Open/Close information for experimentation       |

Robustness experiments showed that the task did not require the full high-dimensional feature set. A compact set of 19 whole-cycle statistics was sufficient to achieve perfect stratified 5-fold validation on the supplied training cycles and near-perfect forward-time validation. **The final 19 features are the mean, standard deviation, minimum, maximum, and range for motor current, motor voltage, electrodynamic force, and door leaf position, with the available electrodynamic-force feature set omitting the redundant minimum term.**

### **Classification and Validation**

Several baseline classifiers were evaluated, including a majority-class dummy classifier, Logistic Regression, Linear SVM, RBF SVM, and Random Forest. All four real classifiers achieved 100% stratified 5-fold out-of-fold accuracy on the 110 labelled cycles, while the dummy baseline achieved 72.7% accuracy and detected no abnormal cycles.

An **additional experiment** compared models with and **without the Open/Close operation indicator. Performance remained unchanged**, indicating that explicit operation context was not necessary for the compact classifier to separate the two classes in the supplied training data.

### **Robustness Validation**

Because 100% cross-validation performance can be misleading on a small dataset, the result was stress-tested with alternative validation schemes. Chronological block

validation, forward-time validation, repeated stratified splits with multiple random seeds, and feature-family ablation were all evaluated.

| Robustness test                  | Observed result                                                         |
|----------------------------------|-------------------------------------------------------------------------|
| Random stratified 5-fold CV      | 100% accuracy, 100% macro F1                                            |
| Chronological block validation   | 100% accuracy across all five blocks                                    |
| Forward-time validation          | Approximately 99.7% mean accuracy;<br>approximately 98.5% minimum split |
| Multiple random seeds            | 100% accuracy across all tested seeds                                   |
| Whole-cycle feature family alone | 100% accuracy                                                           |
| Three selected features          | Approximately 99.1% accuracy                                            |

These results provide strong evidence that the supplied training cycles contain a highly separable fault signature. They do not, however, guarantee 100% performance on the organiser-held test labels as the hidden test distribution remains unseen.

### **Final Model and Test Inference**

The final classifier was trained as a StandardScaler + Logistic Regression pipeline using the 19 whole-cycle features. The complete pipeline was fitted on all 110 labelled training cycles and saved as `door_final_model.joblib`. The exact feature list was saved separately to ensure test-time feature generation matches training.

The final inference notebook applied the same pipeline to the unlabelled `Test.csv` stream. The test file contained 6,253 readings. Timestamp parsing succeeded without invalid values, 37 large timestamp gaps were detected, and therefore 38 candidate door operations were produced. The 19 expected model features were generated without missing or infinite values.

| Test inference check            | Result |
|---------------------------------|--------|
| Test readings                   | 6,253  |
| Large timestamp gaps            | 37     |
| Detected operations             | 38     |
| Invalid segments                | 0      |
| Feature NaNs / infinities       | 0      |
| Normal predictions              | 29     |
| Abnormal resistance predictions | 9      |
| Output rows                     | 38     |

The generated `door_predictions.csv` follows the required Door submission schema with one row per predicted segment and the columns `start_time`, `end_time`, and `prediction`. The inference notebook also checks for valid labels, missing values, duplicate rows, and invalid temporal intervals before saving the final file.